

Piyush Hinduja

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OBJECTIVE

Data-Driven Machine Learning Engineer with expertise in NLP, LLMs, GenAI, Deep Learning, and scalable data processing. Skilled in architecting high-performance predictive models and leveraging AI to extract actionable insights. Passionate about implementing results-driven, decision-making and enhancing business intelligence through cutting-edge ML solutions.

EDUCATION

Master of Science in Computer Science <i>University of Utah</i>	GPA: 3.9/4.0 2023 – 2025
Bachelor of Engineering in Computer Engineering <i>University of Mumbai</i>	GPA: 3.85/4.0 2019 – 2023

SKILLS

Languages: *Expert (5+ yrs):* Python, Java, JavaScript *Intermediate (3+ yrs):* C, C++ *Beginner (1+ yrs):* Kotlin, SQL, Shell
Machine Learning: LLMs, PyTorch, TensorFlow, Scikit-Learn, BERT models, EasyOCR, HuggingFace, Langchain
Data Processing: RESTful APIs, Pandas, NumPy, PySpark, Databricks, OCR Pipelines, Matplotlib, Seaborn
Applications and Databases: Streamlit, Bash, Preswald, MongoDB, MySQL, Amazon RDS, Firebase, ChromaDB
Cloud & Miscellaneous: AWS (EC2, S3, RDS), GCP, Azure, Docker, Kubernetes, CloudLab, Git, CI/CD, Agile/SCRUM, JIRA

WORK EXPERIENCE

Data Analyst <i>University of Utah</i>	March 2024 – Present <i>Salt Lake City, Utah</i>
- Implemented advanced NLP analysis using encoder-only model FinBERT, PyTorch, and Transformers for financial document processing, improving the prediction accuracy from 66% to 94.13% - Reduced per-filing processing time by 75% using multi-cores and multi-thread environments to parallelize file processing, API requests, and text cleaning operations - Developed robust prompt engineering solutions to integrate LLMs & GenAIs (OpenAI, Anthropic, and Gemini) for automated financial text analysis, comparing AI-generated insights against traditional algorithms - Optimized financial data processing workflows, achieving faster execution and lower memory consumption through optimized data structures and parallelized computations	
Machine Learning Researcher <i>FLUX Research Group</i>	August 2024 – Present <i>Salt Lake City, Utah</i>
- Engineered automated deployment system using Python and Bash to streamline NVIDIA/CUDA toolkit integration on high-performance computing clusters, cutting setup time by 90% for 150+ researchers - Implementing time-series forecasting model to analyze the campus's wireless spectrum data, optimizing power distribution across 100+ routing devices on campus	
Full Stack Developer Intern <i>Exposys Data Labs</i>	June 2021 – July 2021 <i>Bangalore, India</i>
- Spearheaded the launch of a tourism project, integrating key features to enhance user experience and deliver valuable information about 50+ Indian travel destinations - Utilized MERN stack for an efficient frontend and backend integration, creating a responsive, user-friendly interface and enabling content management	

PROJECTS

SEC Comment Letter Analysis <i>LLMs, OCR, Claude API</i>	
- Engineered a rolling-window algorithm to process and bundle SEC comment letter conversations, analyzing data from 50,000+ companies via the EDGAR database - Designed an OCR pipeline using the EasyOCR library to extract text from PDFs with 95% accuracy, processing 2-page documents in under 2 seconds - Constructed prompt-based labeling system using Claude API to classify letter content with 98.25% accuracy, integrating metadata for contextual inference	
Road Damage Detection <i>Object-Detection, YOLOv5, FasterRCNN</i>	GitHub Link
- Collaborated with a team of four to develop a road damage detection system using YOLOv5 and FasterRCNN, integrating a user-friendly web interface (deployed via Streamlit), leveraging the RDD-2020 dataset - Achieved model accuracies of 52.67% with YOLOv5 and 44.45% with FasterRCNN by training and fine-tuning on different hyper-parameters - The system plays a vital role, delivering a 70% improvement in road damage detection accuracy over manual methods, helping lower the frequency of accidents and fatalities	
Dependency Parser <i>PyTorch, GloVe Embeddings</i>	GitHub Link
- Architected an end-to-end Transition-Based Dependency Parser in PyTorch, implementing a multi-class classifier for predicting parsing actions based on stack and buffer states - Designed modular pipeline incorporating data preprocessing, model architecture, training workflow, and evaluation metrics - Integrated GloVe word embeddings for enhanced semantic representation, contributing to UAS of 0.76 and LAS of 0.705	
Language Modeling <i>NLP, RNN, N-gram, PyTorch</i>	GitHub Link
- Developed character-level LSTM model in PyTorch to predict character sequences based on 50-character context windows - Implemented teacher forcing during training to prevent error propagation, using ground truth characters instead of predictions for subsequent timesteps - Achieved perplexity of 9.43 through systematic model architecture optimization and training methodology	